

8] Comparison - element wise Comparison.

list1 = [1, 2, 3, 4]

list2 = [1, 2, 5, 6]

list1 < list2
3 < 5

print (list1 < list2) → True

print (list1 > list2) → False

9] Emptiness - not operator

eg. list1 = []

print (not list1) → True

eg. list2 = [10, 20, 30]

print (not list2) → False

eg. list3 = [] → True/False

if not list3:

print ("list is empty")

else :

print("list is not empty")

- Built in functions in list.

1] len() - len() function prints

Count of element in the list

eg. list1 = [10, 20, 30, 40, 50]

1 2 3 4 5

print(len(list1)) → 5

Built-in function Syntax

fun_name(list_name)

len(list1)

2] max() - max() function

prints largest element in the list

eg. list1 = [50, 30, 20, 10, 40]

print(max(list1)) → 50

3] `min()` - `min()` function

prints the smallest element
in the list.

eg. `list1 = [50, 40, 10, 20, 30]`

`print(min(list1))` → 10

4] `Sum()` - `Sum()` function

prints/returns, sum of all elements
in the list.

eg. `list4 = [10, 20, 30, 40, 50]`

`print(sum(list4))` → 150

5] `Any()` - `any()` function returns

true, if at least one element
of the list is true.

eg. `list1 = [0, 0, 0, 0, 1]`

0 & False $\Rightarrow$ False

1, 100, 10.1, 1+2j, "ganesh" $\Rightarrow$ True

print(any(list1)) $\rightarrow$ True

eg. list2 = [0, 0, 0, 0, 0]

F F F F F

print(any(list2)) $\rightarrow$ False

6] all() - all() function

returns true, if all elements are true

eg. list1 = [10, 20, 30, 1+2j, "ganesh"]

T T T T T

print(all(list1)) $\rightarrow$ True

eg. list2 = [10, 20, 30, 1+2j, 0]

T T F T F

print(all(list2)) $\rightarrow$ False

7] Sorted() - Sorted() function

returns Sorted list.

ex. list 1 = [10, 20, 50, 30, 40]

```
print(sorted(l1st))
```

o/p - [10, 20, 30, 40, 50]
asc ord (sorted)

asc ord (sorted)

8] reversed() - reversed()

function returning list in reversed order.

eg. list1 = [10, 20, 30, 40, 50]

```
list2 = reversed(list1)
```

print(list2) → obj

`print(list(list2))` → list

o/p - [50, 40, 30, 20, 10]

```
print(list(reversed(list)))
```

✓ `max(list)`

- List Methods :-

Syntax :-

`list-name.method-name()`

eg. `list1.append()`

- Built in function :-

Syntax :-

`function-name(list-name)`

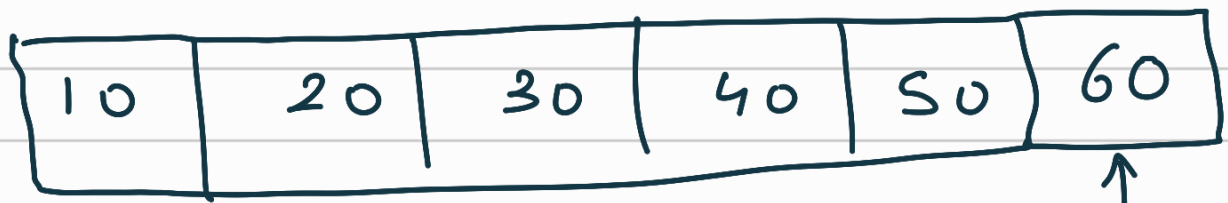
eg. `len(list1)`

List Methods :-

[1] Adding Elements in the list

1] `append()` — Using this method, we can add element at the end of the list

eg. list1 = [10, 20, 30, 40, 50]

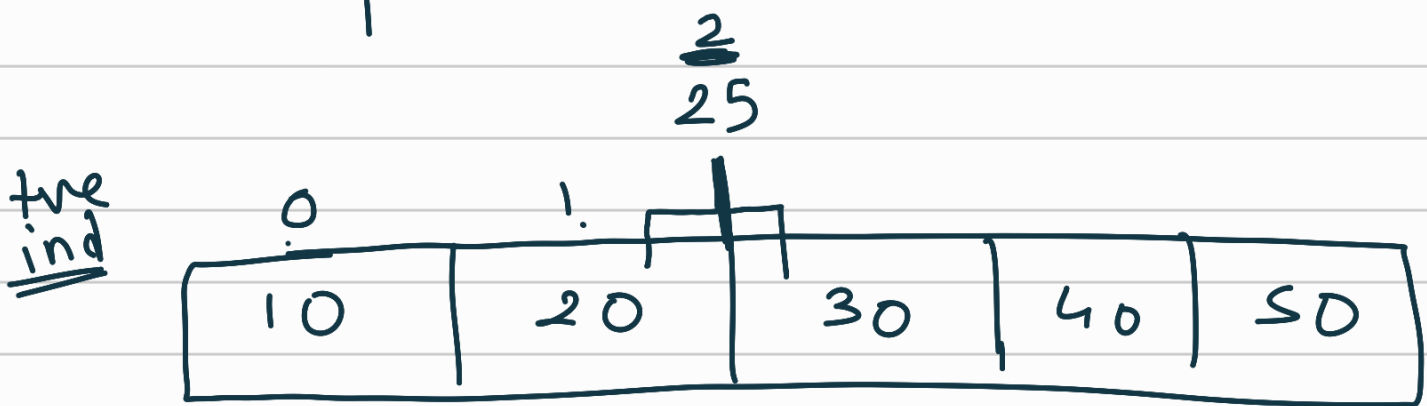


↑
add element
at the end

list1.append(60)

print(list1) → [10, 20, 30, 40, 50, 60]

2] insert() — Using insert method, we can add/insert element at the particular index or position.



[10, 20, 25, 30, 40, 50]

list1 = [10, 20, 30, 40, 50]

list1.insert(2, 25)

`print(list1)` $\Rightarrow$ ^{position} `[10, 20, 25, 30, 40, 50]` ^{element}

3] `extend()` - Using `extend` method, we can add one list at the end of another list.

eg, `list1 = [10, 20, 30]`
`list2 = [40, 50, 60]`

`list1.extend(list2)`
`print(list1)`

o/p- `[10, 20, 30, 40, 50, 60]`

2] removing elements from the list.

1] `remove()` - Using `remove`

method, we can remove particular element from the list.

eg. list1 = [10, 20, ~~30~~, 40, 50]

list1.remove(30)

print(list1) → [10, 20, 40, 50]

2] pop() - we can use pop() method in two ways

i) first way - pop() - It will remove last element of the list

eg. list1 = [10, 20, 30, 40, 50]

list1.pop()

print(list1) → [10, 20, 30, 40]

ii) Second way - pop(index)

when we provide index no. to this method, it will remove element at that specific index.

eq. list1 = [10, 20, 30, 40, 50]

list1.pop(2)

print(list1) → [10, 20, 40, 50]

7] index() - Using index() method

we can print index of that particular element.

eq. list1 = [10, 20, 30, 40, 50]

index = list1.index(40)

print(index) → 3

print(list1.index(30))
→ 2

8) `count()` - Using `count()` method, we can print no. of times particular element appear in the list.

eg. `list1 = [10, 20, 30, 10, 50, 20, 10]`
 ① ① ② ② ③

`print(list1.count(10))` → 3

`print(list1.count(20))` → 2

Built-in function

`sorted()` -

`reversed()` -

• method

`sort()` -

`reverse()` -

9) `sort()` - Using `sort` method we can rearrange the

elements of the list in ascending order.

eg. list1 = [50, 40, 30, 5, 45, 35, 10]

```
list1.sort()
```

```
print(list1)
```

→ [5, 10, 30, 35, 40, 45, 50]

10] reverse() - Using

reverse() method, we can print the elements of the list in reverse order.

eg. list1 = [10, 20, 30, 40, 50]

```
list1.reverse()
```

```
print(list1)
```

o/p - [50, 40, 30, 20, 10]

11] `copy()` - Using `copy()` method, we can create a copy of the list.

eg. `list1 = [10, 20, 30, 40, 50]`

`list2 = list1.copy()`

`print(list2)`

o/p - `[10, 20, 30, 40, 50]`